

PYTHON Problem Set

PYTHON Design Problems

1. FIFO Queue Class

- File: `fifo\_queue`
- Description: Enqueue/dequeue/peek

2. Palindrome Checker

- File: `palindrome`
- Description: String same forwards/backwards

3. Fibonacci Generator

- File: `fib\_gen`
- Description: Iterator with yield

4. Dictionary Frequency Counter

- File: `freq\_counter`
- Description: Word counts

5. Merge Two Sorted Lists

- File: `merge\_sorted`
- Description: Sorted output

6. LRU Cache

- File: `lru\_cache`
- Description: Use OrderedDict or custom

7. Producer-Consumer

- File: `prod\_cons`
- Description: Threading + Queue

8. Matrix Rotation 90°

- File: `matrix\_rot`
- Description: Rotate NxN matrix

9. QuickSort

- File: `quicksort`
- Description: Recursive divide & conquer

10. Async Web Scraper

- File: `async\_scraper`
- Description: aiohttp multiple URLs